LLAVA AND PLLAVA

- 汇报人：张景深

论文分享流程

一、LLaVA

二、PLLaVA

一、LLAVA

文章试图解决的问题：
提高多模态领域大语言模型的零样本能力

- 多模态指令跟随数据生成
- 多模态大模型框架的设计
- 多模态指令跟随基准测试

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], person: [0.63, 0.222, 0.686, 0.516], person: [0.444, 0.233, 0.487, 0.34], backpack: [0.384, 0.696, 0.485, 0.914], backpack: [0.755, 0.413, 0.846, 0.692], suitcase: [0.758, 0.413, 0.845, 0.69], suitcase: [0.1, 0.497, 0.173, 0.579], bicycle: [0.282, 0.363, 0.327, 0.442], car: [0.786, 0.25, 0.848, 0.322], car: [0.783, 0.27, 0.827, 0.335], car: [0.86, 0.254, 0.891, 0.3], car: [0.261, 0.101, 0.787, 0.626]

Response type 1: conversation

Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV).
Question: Where is the vehicle parked?
Answer: The vehicle is parked in an underground parking area, likely in a public garage.
Question: What are the people in the image doing?
Answer: The people are trying to fit all of their luggage into the SUV, likely preparing for a trip.

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip.
In the parking area, various luggage items are scattered around the vehicle. There are two backpacks, one located near the left rear wheel and the other closer to the right side of the vehicle. Additionally, there are two suitcases, one on the right side of the car and another further away near the center of the parking area. A bicycle can also be seen on the left side of the vehicle.
Other cars are parked around the main SUV, with one car positioned behind it and slightly to the left, another behind and slightly to the right, and the third car further behind on the right side.

Response type 3: complex reasoning

Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings to accommodate. They might have to strategize and arrange the luggage efficiently to ensure that everything fits properly into the vehicle. Additionally, they need to consider the comfort of the passengers and visibility while driving, so the placement of the luggage must not obstruct the driver's view or make the passengers uncomfortable during the trip.

Table 14: One example to illustrate the instruction-following data. The top block shows the contexts such as captions and boxes used to prompt GPT, and the bottom block shows the three types of responses. Note that the visual image is not used to prompt GPT, we only show it here as a reference.

文章提出的策略：

- **1、多模态指令跟随数据合成：**

使用GPT-4或ChatGPT生成多模态指令跟随数据。首先，将图像转换为文本描述（如标题和边界框），然后生成一系列问题来指导助手描述图片内容。

文章提出的策略：

- **2、多模态模型框架：**

连接CLIP的视觉编码器和Vicuna语言解码器，并通过一个可训练的投影矩阵将视觉特征映射到语言模型的词嵌入空间。

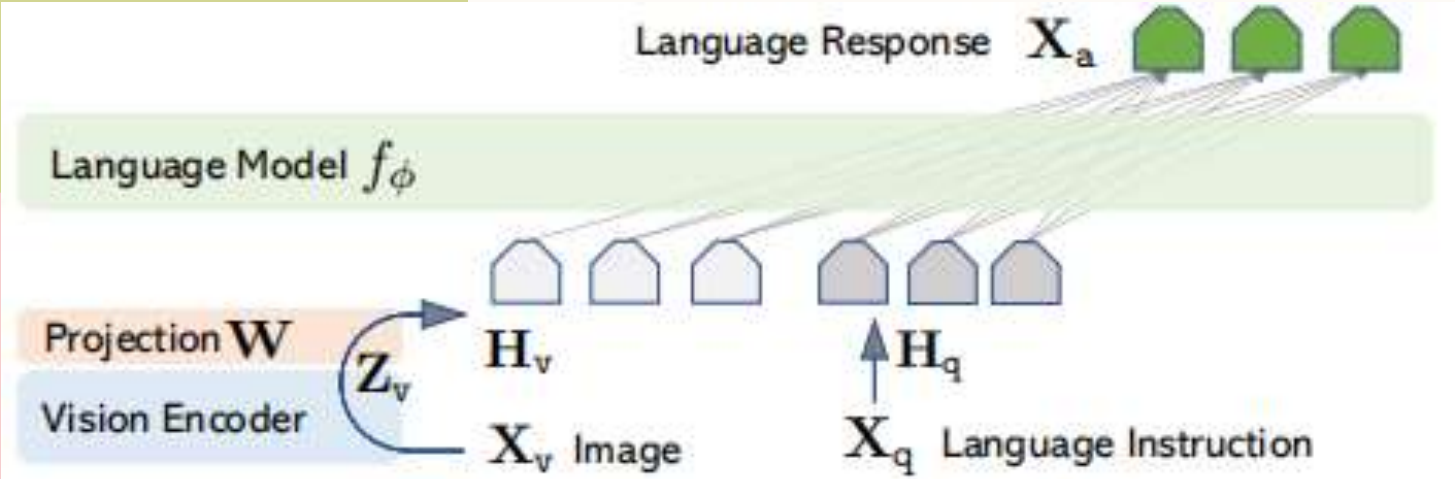


Figure 1: LLaVA network architecture.

文章提出的策略：

- **3、多阶段训练：**
  - **预训练实现特征对齐**

    使用过滤后的CC-595K图像-文本对，将图像转换为指令跟随数据，并通过最大化似然来训练投影矩阵，使图像特征与预训练的语言模型词嵌入对齐。

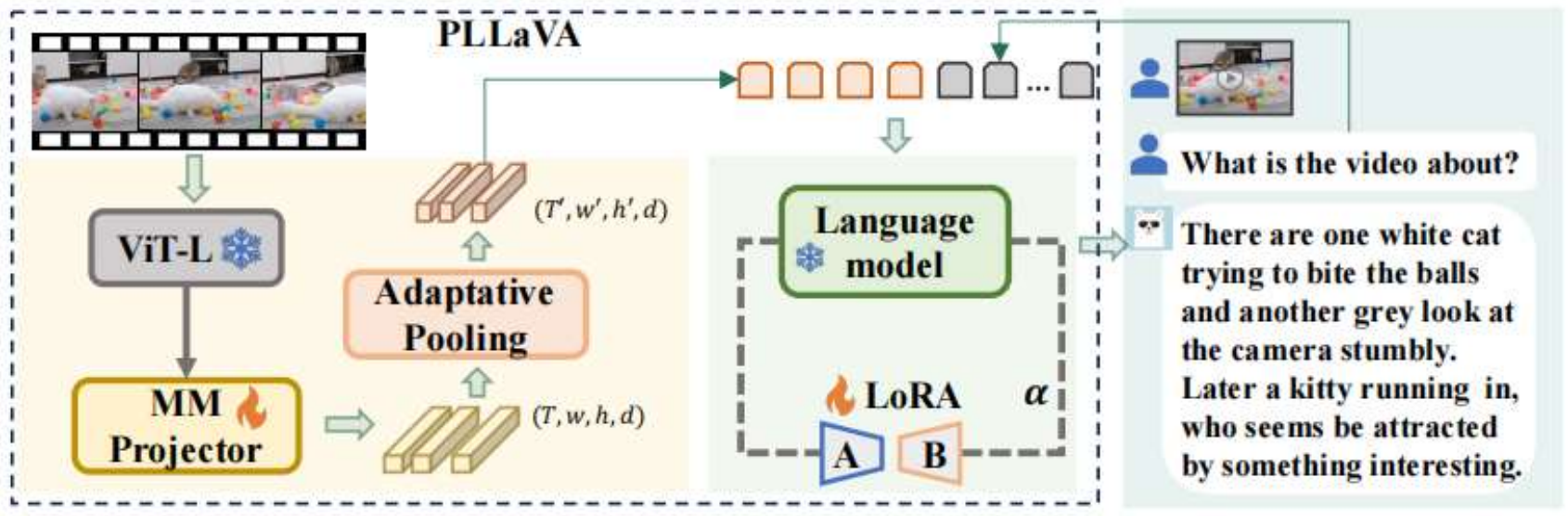
  - **端到端微调**

    保持视觉编码器权重固定，更新投影层和语言模型的预训练权重，以进一步提高模型在多模态任务上的性能。

二、 PLLAVA

文章试图解决的问题：将现有的图像-语言预训练模型用于视频理解任务。

- 适应性池化策略
- 后训练优化
- 资源轻量化



文章的动机：

- 1、视频相关任务的预训练过程计算与数据资源消耗大；
- 2、直接微调预训练图像语言模型，将多帧作为视频数据集输入会导致模型性能饱和甚至下降。

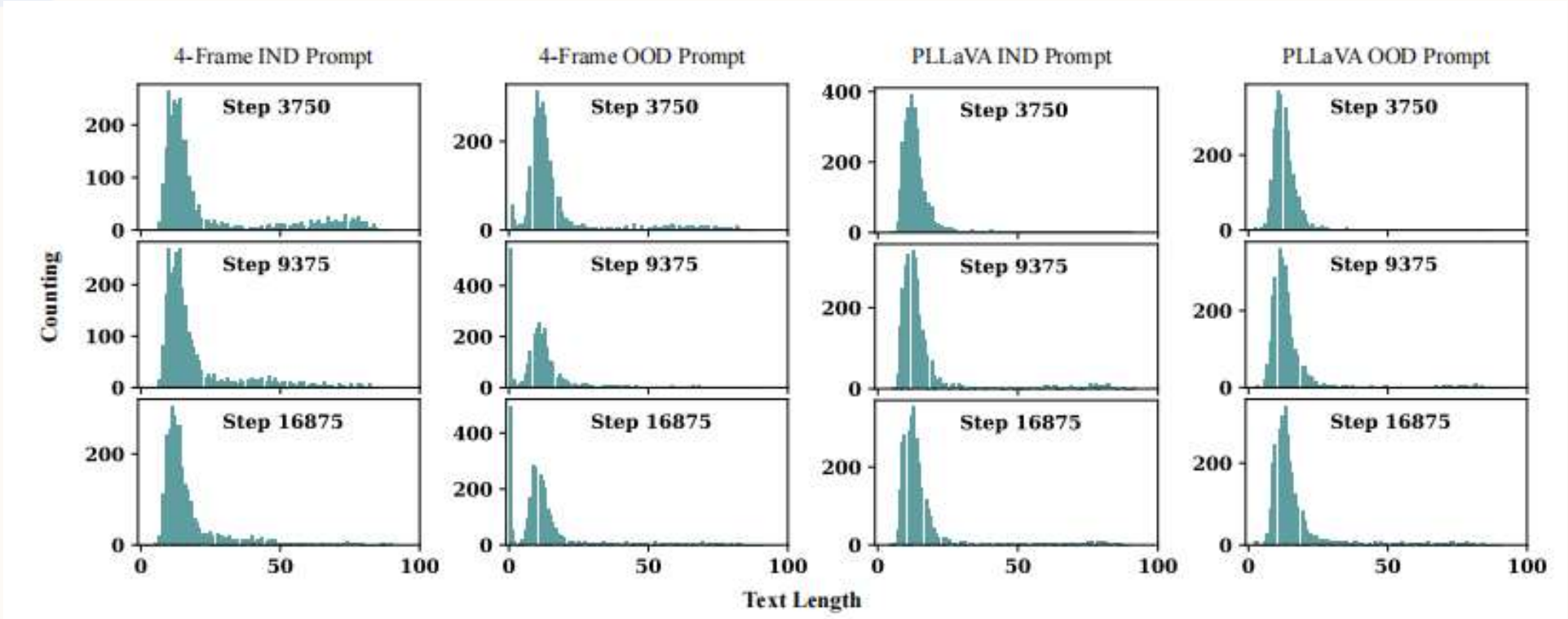
先前策略：

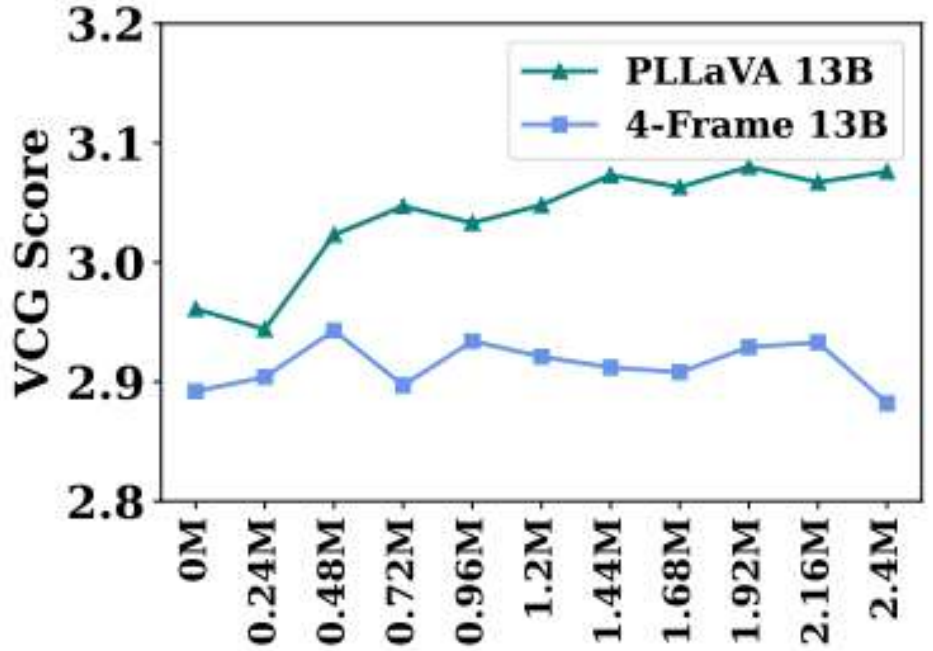
- 将多个视频帧编码为一系列特征，以序列形式直接馈送给模型
- 不足：

  （1）与零样本相比，模型并不总能提高性能，反而对prompt的变化引入性能漏洞

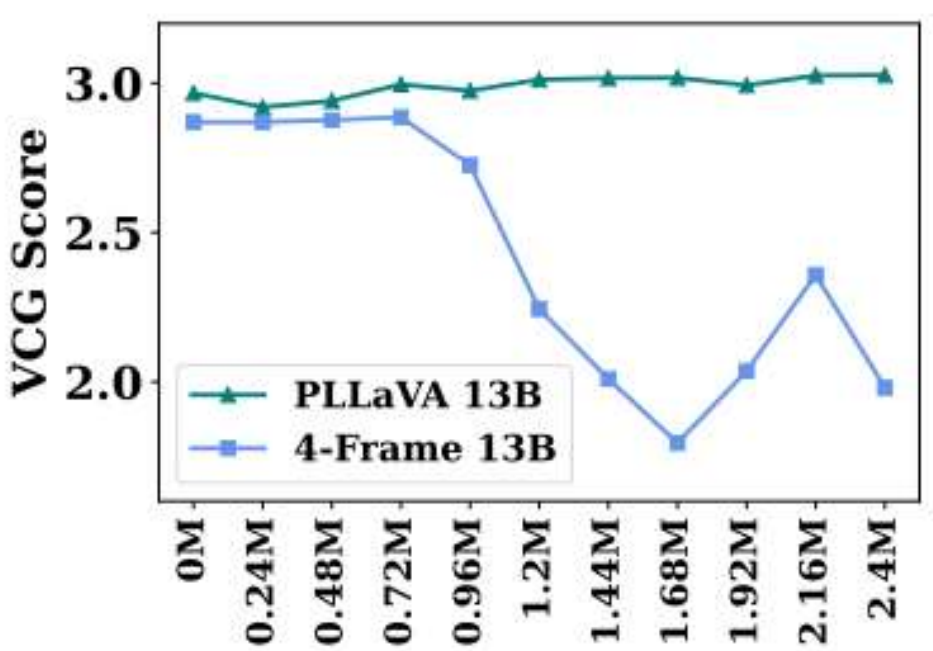
  （2）扩大模型尺寸并不会提高模型的推理性能

- **文本长度直方图：**
  - **4-Frame随训练步长增加，产生长文本生成遗忘**
  - **PLLaMA保持一致**





(a) IND prompt.

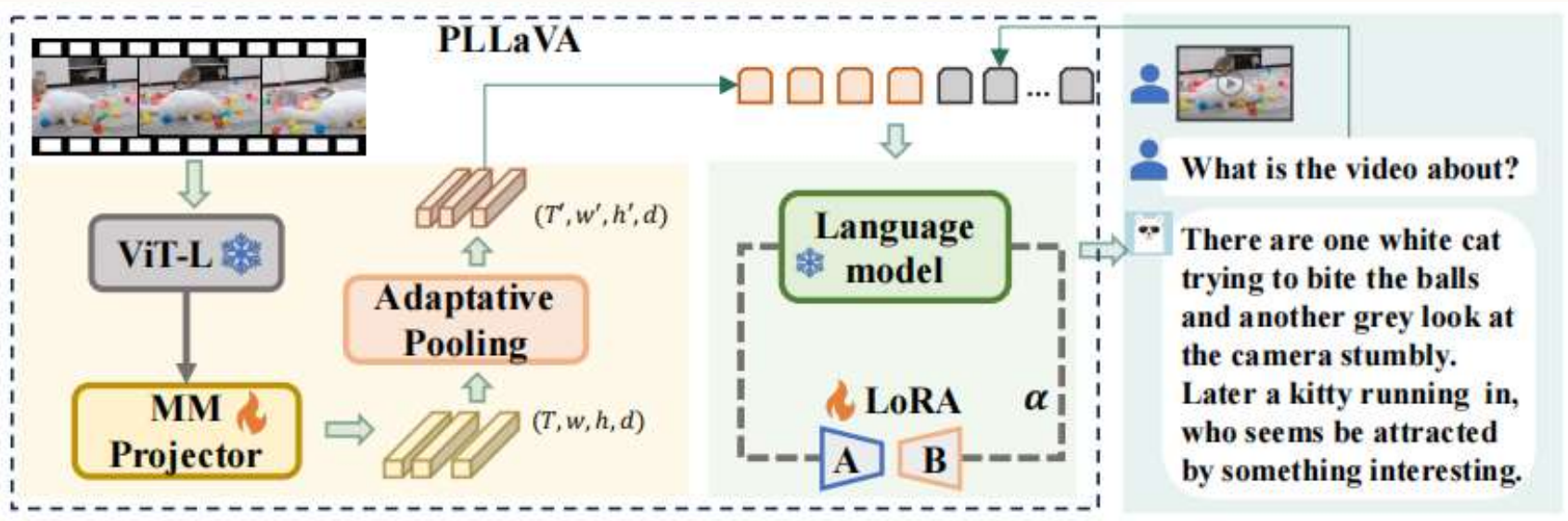


(b) OOD prompt.

文章提出的策略：

- 1、适应性池化策略：

为了在保留时间信息的同时减少计算成本，PLLaVA采用了一种参数自由的适应性池化操作来压缩视频特征的空间和时间维度。这有助于平衡训练效率和描述准确性。

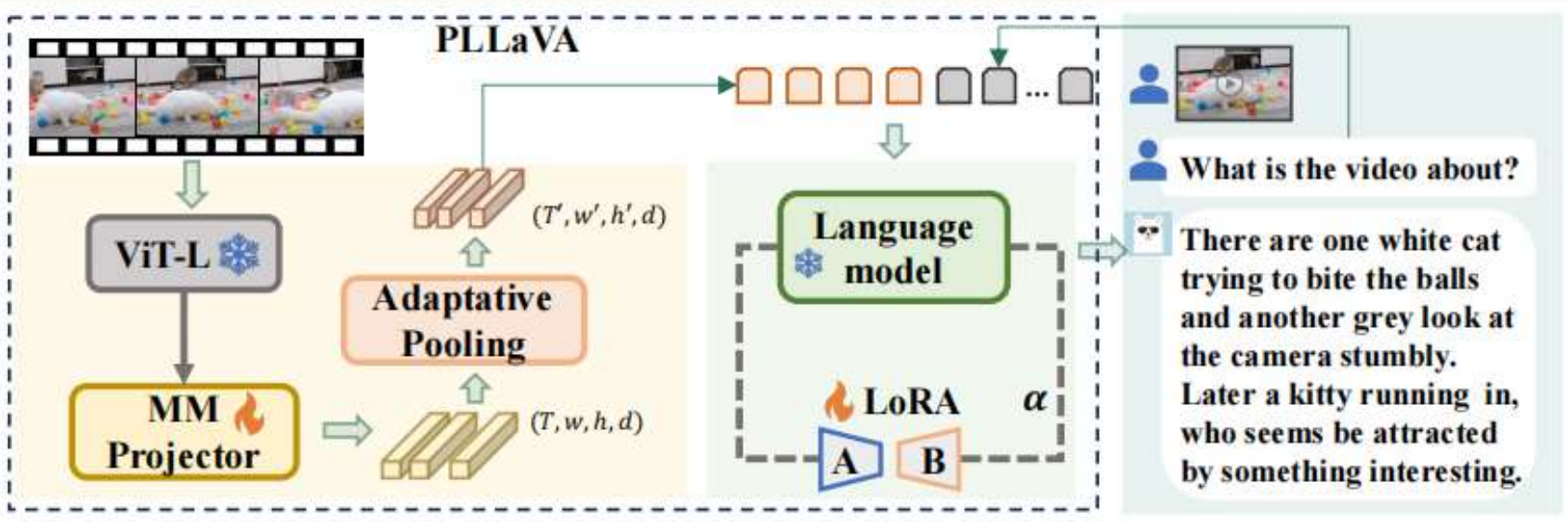


文章提出的策略：

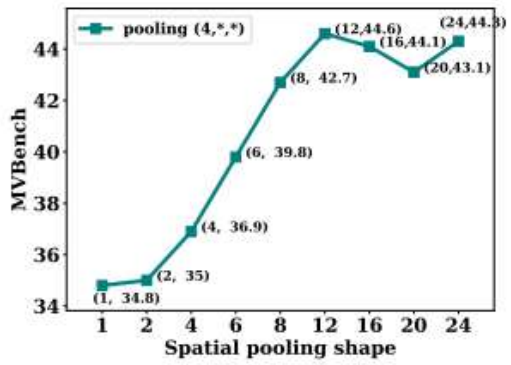
- 2、后训练优化：

为了解决大型语言模型在视频数据集上微调时性能下降的问题，论文提出了一种后训练优化方法。这种方法通过权重融合技术结合了图像预训练模型权重和视频数据集微调后的权重，以减少在多模态微调过程中的遗忘现象。

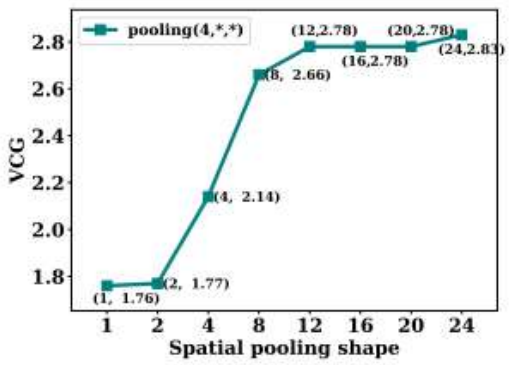
- 性能下降可能源于对低质量视频文本数据样本进行训练而导致的语言熟练程度降低。



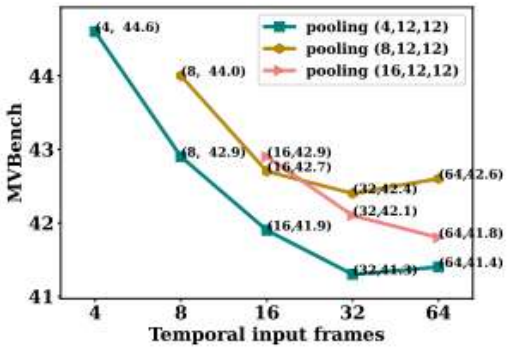
（4，*，*）表示空间维度



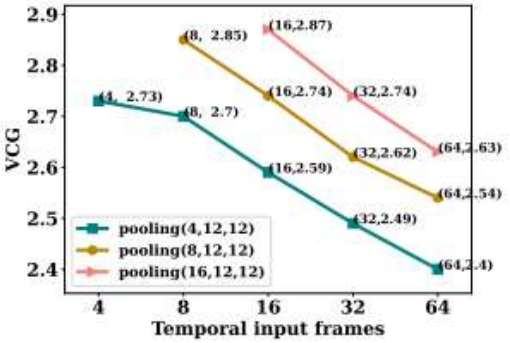
(a) Spatial shape effects on MVBench.



(b) Spatial shape effects on VCG.



(c) Temporal shape effects on MVBench.



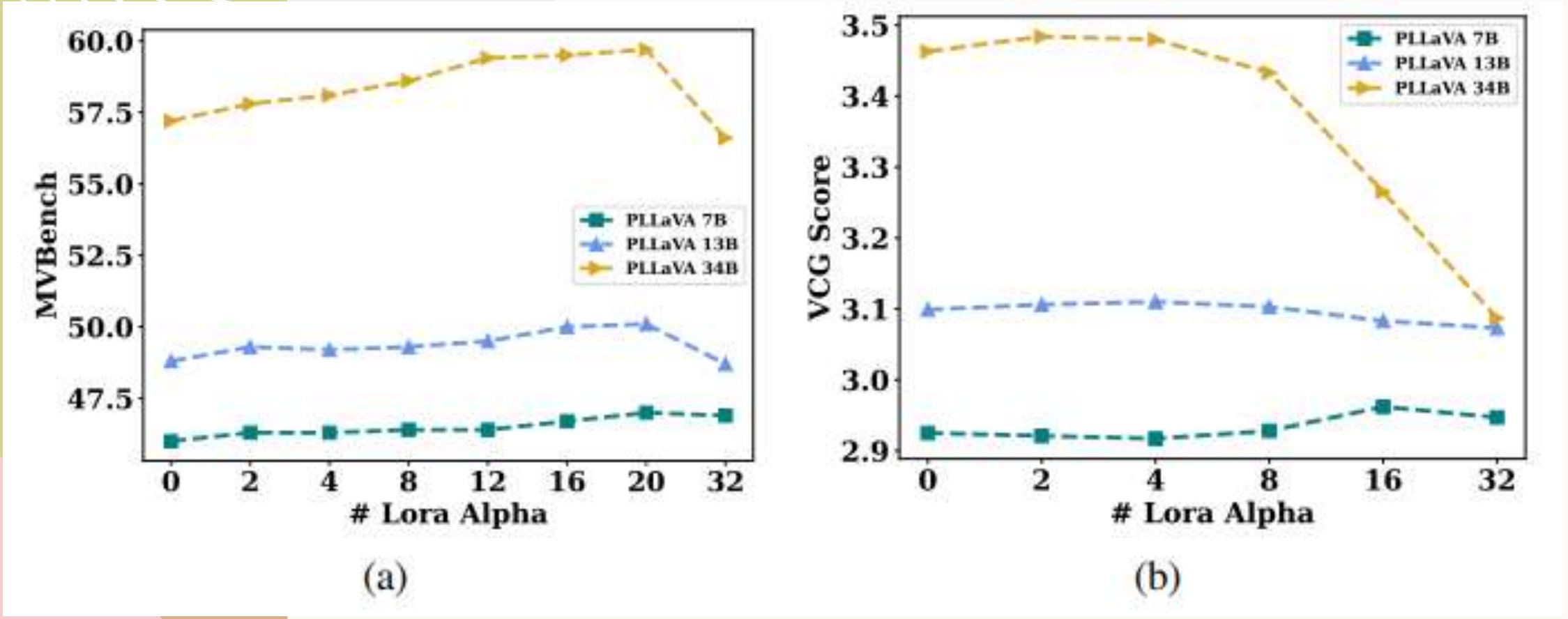
(d) Temporal shape effects on VCG.

结果：

- **1、池化操作的设计：**
  - 哪个维度更适合合并？
  - 沿着这个维度最大压缩比？

（MVBench：多模态大模型视频理解能力基准）

（时间维度）改变视频帧输入数量，时间池化敏感，沿着时间池化总会降低模型性能


MVBench
60.0
57.5
55.0
52.5
50.0
47.5
0
2
4
8
12
16
20
32
Lora Alpha
PLLaVA 7B
PLLaVA 13B
PLLaVA 34B
(a)
VCG Score
3.5
3.4
3.3
3.2
3.1
3.0
2.9
0
2
4
8
16
32
Lora Alpha
PLLaVA 7B
PLLaVA 13B
PLLaVA 34B
(b)

Method	Vision Encoder	LLM Size	MSVD-QA		MSRVTT-QA		ActivityNet-QA		TGIF-QA		Video-ChatGPT					
			Acc.	Sco.	Acc.	Sco.	Acc.	Sco.	Acc.	Sco.	CI	DO	CU	TU	CO	Avg.
FrozenBiLM[45]	ViT-L	1.3B	33.8	-	16.7	-	25.9	-	41.9	-						
Video-LLaMA[50]	CLIP-G	7B	51.6	2.5	29.6	1.8	12.4	1.1	-	-	1.96	2.18	2.16	1.82	1.79	1.98
LLaMA-Adapter[51]	ViT-B	7B	54.9	3.1	43.8	2.7	34.2	2.7	-	-	2.03	2.32	2.30	1.98	2.15	2.16
Video-ChatGPT[33]	ViT-L	7B	64.9	3.3	49.3	2.8	35.2	2.7	51.4	3.0	2.50	2.57	2.69	2.16	2.20	2.42
Video-LLaVA[25]	ViT-L	7B	70.7	3.9	59.2	3.5	45.3	3.3	70.0	4.0						
Chat-UniVi[14]	ViT-L	7B	65.0	3.6	54.6	3.1	45.8	3.2	60.3	3.4	2.89	2.91	3.46	2.89	2.81	2.99
MovieChat[37]	CLIP-G	7B	75.2	3.8	52.7	2.6	45.7	3.4	-	-	2.76	2.93	3.01	2.24	2.42	2.67
VideoChat[19]	CLIP-G	7B	56.3	2.8	45.0	2.5	26.5	2.2	34.4	2.3	2.23	2.50	2.53	1.94	2.24	2.29
VideoChat2[20]	UMT-L	7B	70.0	3.9	54.1	3.3	49.1	3.3	-	-	3.02	2.88	3.51	2.66	2.81	2.98
Vista-LLaMA[32]	CLIP-G	7B	65.3	3.6	60.5	3.3	48.3	3.3	-	-	2.44	2.64	3.18	2.26	2.31	2.57
LLaMA-VID[21]	CLIP-G	13B	70.0	3.7	58.9	3.3	47.5	3.3	-	-	2.96	3.00	3.53	2.46	2.51	2.89
LITA [13]	CLIP-L	7B	-	-	-	-	-	-	-	-	2.94	2.98	3.43	2.68	3.19	3.04
ST-LLM [31]	BLIP2	7B	74.6	3.9	63.2	3.4	50.9	3.3	-	-	3.23	3.05	3.74	2.93	2.81	3.15
IG-VLM CogAgent[10]	CLIP-E	7B	76.7	4.1	62.7	3.6	57.3	3.6	76.7	4.0	3.26	2.76	3.57	2.34	3.28	3.04
IG-VLM LLaVA 7B [28]	ViT-L	7B	78.8	4.1	63.7	3.5	54.3	3.4	73.0	4.0	3.11	2.78	3.51	2.44	3.29	3.03
IG-VLM LLaVA 13B [28]	ViT-L	13B	77.4	4.1	62.6	3.4	57.1	3.5	78.0	4.0	3.17	2.79	3.52	2.51	3.25	3.05
IG-VLM LLaVA 34B [28]	ViT-L	34B	79.6	4.1	62.4	3.5	58.4	3.5	79.1	4.2	3.21	2.87	3.54	2.51	**3.34**	3.09
IG-VLM GPT-4V[1]	Unk	GPT-4	76.3	4.0	63.8	3.5	57.0	3.5	65.3	3.7	3.40	2.80	3.61	**2.89**	3.13	3.17
PLLaVA 7B	ViT-L	7B	76.6	4.1	62.0	3.5	56.3	3.5	77.5	4.1	3.21	2.86	3.62	2.33	2.93	3.12
PLLaVA 13B	ViT-L	13B	75.7	4.1	63.2	3.6	56.3	3.6	77.8	4.2	3.27	2.99	3.66	2.47	3.09	3.27
PLLaVA 34B	ViT-L	34B	**79.9**	**4.2**	**68.7**	**3.8**	**60.9**	**3.7**	**80.6**	**4.3**	**3.60**	**3.20**	**3.90**	2.67	3.25	**3.48**
Improve over GPT-4V [16]	-	-	+3.6	+0.2	4.9	0.3	3.9	0.2	15.3	0.6	0.2	0.4	0.3	-0.32	0.12	0.31

Table 2: Results of video question-answering.

总结：

- 池化策略?
- 视觉编码器?

谢谢观看！

汇报人：张景深